Q1
THE CRATE HAS WEIGHT DOWN, THE NORMAL FORCE ON OF THE RAMP, AND FRICTION UP THE SLOPE, BECAUSE NOTHING ELSE IS TOUCHING IT
THE NORMAL FORCE IS PERPENDICULAR TO THE RAMP AND FRICTION IS PARALLEL TO IT.
THE CRATE STAYS PUT BECAUSE FRICTION BALANCES THE WEIGHT COMPONENT ALONG THE RAMP, BECAUSE THE [illegible] FORCE ARE EQUAL AND OPPOSITE.

O
USING EQUAL ARCS FROM A AND FROM B
AND JOINED THE ARC CROSSINGS BECAUSE
THE CROSSINGS ARE EQUIDISTANT FROM BOTH
END POINTS.
THE ARCS MEET AT P AND Q AND PA EQUALS
PB AND QA EQUALS QB, BECAUSE EACH ARC
WAS DRAWN WITH ONE RADIUS, AND THIS
HOLDS AS LONG AS THE COMPASS IS NOT
RESET BETWEEN ARCS.
PQ IS PERPENDICULAR TO AB AND CUTS IT
IN HALF BECAUSE P AND Q ARE BOTH
EQUIDISTANT FROM A AND B.

Q3
THE HORIZONTAL AXIS IS TIME IN SECONDS
AND THE VERTICAL AXIS IS VELOCITY IN
METRES PER SECOND, BECAUSE THE AXIS
LABELS CARRY THEIR UNITS. AND THIS
HOLDS AS LONG AS THE SCALE IS LINEAR ON
BOTH AXES.
IT SPEEDS UP FOR FOUR SECONDS, TURNS
AROUND AT THE PEAK, THEN SLOWS TO REST.
THE DISPLACEMENT OVER THE FIRST FOUR
SECONDS IS THE AREA UNDER THE CURVE.
SIXTEEN METRES, BECAUSE AREA UNDER A
VELOCITY-TIME GRAPH IS DISPLACEMENT.

Q4
ALL THREE TRIALS ARE ROUNDED TO ONE DECIMAL PLACE.
THE TABLE SHOWS THE TIME FALLING AS THE RAMP ANGLE RISES. BECAUSE A STEEPER RAMP GIVES A LARGER ALONG-RAMP FORCE. TRIAL TWO SITS OFF THE PATTERN AND NEEDS REPEATING, BECAUSE ITS VALUE IS FAR FROM THE OTHER TWO AT THE SAME ANGLE.
Q5
C-13.
C-14: A
C-15: A